

Microservices Architecture using ASP.NET Core Web API

EXERCISE 1: JWT Authentication & Login :

Credentials Used:

- Username: admin
- Password: admin123
- Role: Admin

Expected Response: 200 OK

Response Contains:

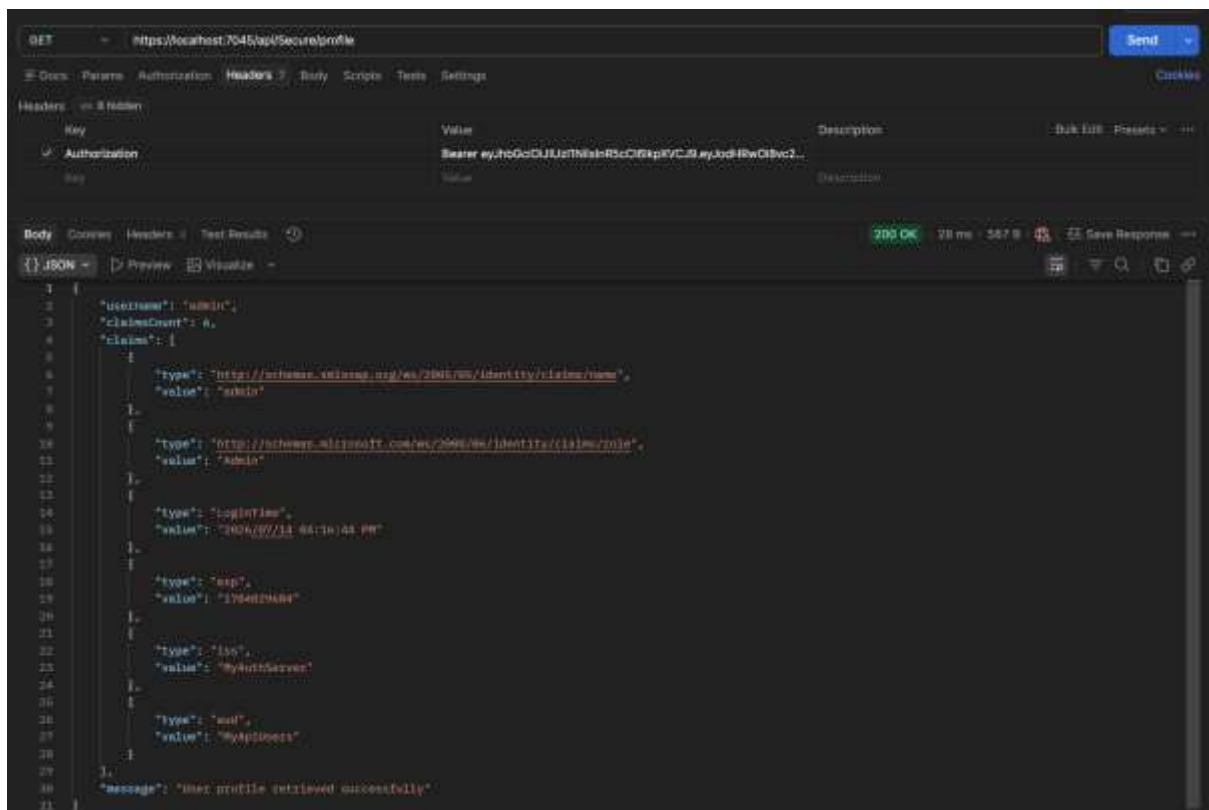
- Valid JWT token
- Message: "Login successful"
- Token expiry: 60 minutes

Token Received:

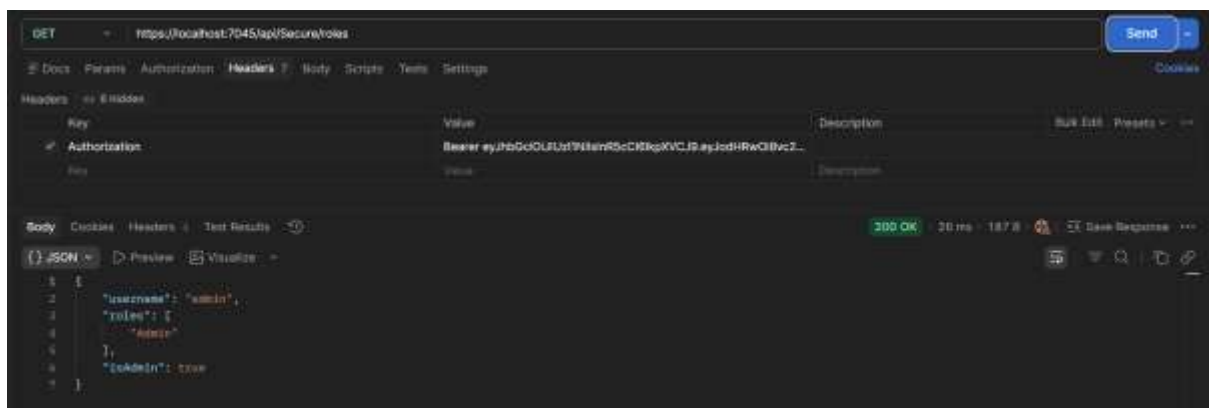
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJodHRwOi8vc2NoZW1hcy54bWxzY2FwLm9yZy93cy8yMDA1LzA1L2lkZW50aXR5L2NsYWltcy9uYWw1IlJoIjEWRtaW4iLCJodHRwOi8vc2NoZW1hcy5taWNyb3NvZnQuY29tL3dzLzlwMDgvdjYvaWRlbnRpdHkvY2xhaW1zL3JvbGUuOjEJBZG1pbilSlkxvZ2luVGltZSI6IjIwMjYyMTQgMDQ6MDU6MzMgUE0iLCJleHAiOiE3ODQwMjg5MzMsImVycyI6IjE5QXV0aFNlcmlzIiwiaWF0IjE1ZXQxNjBpVXNlenMifQ.kPOz5B0gP9INeyYcHI4Uhc_A0508KzaP6QXWb3PPkUc



Test 3: Get User Profile –

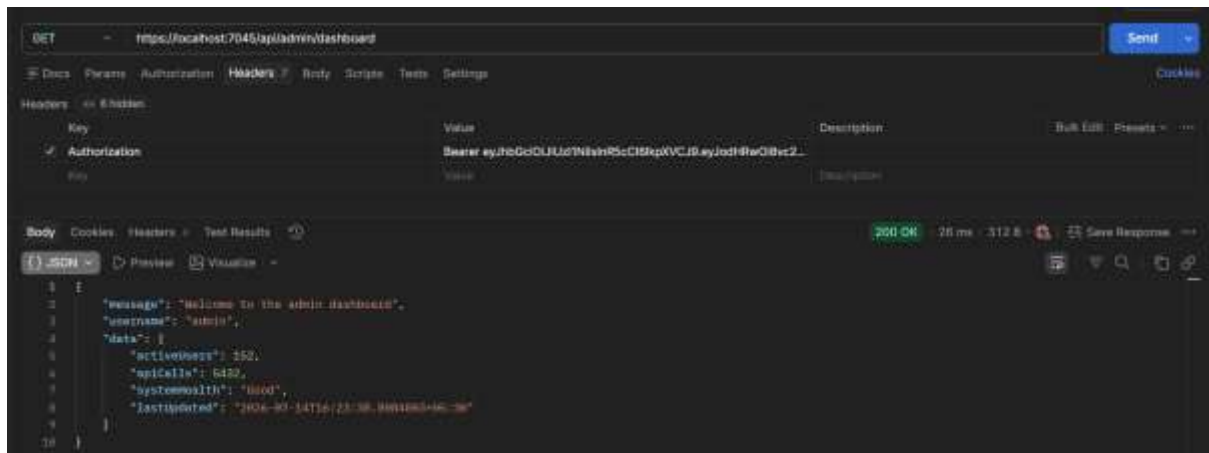


Test 4: Get User Roles –

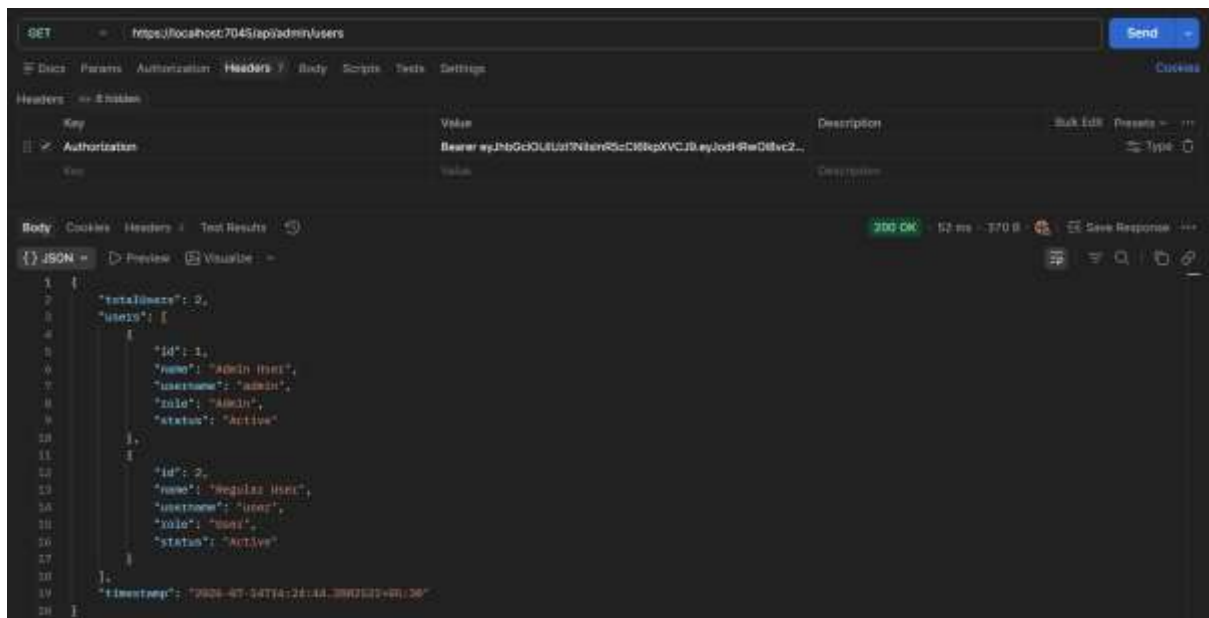


EXERCISE 3: ROLE-BASED AUTHORIZATION :

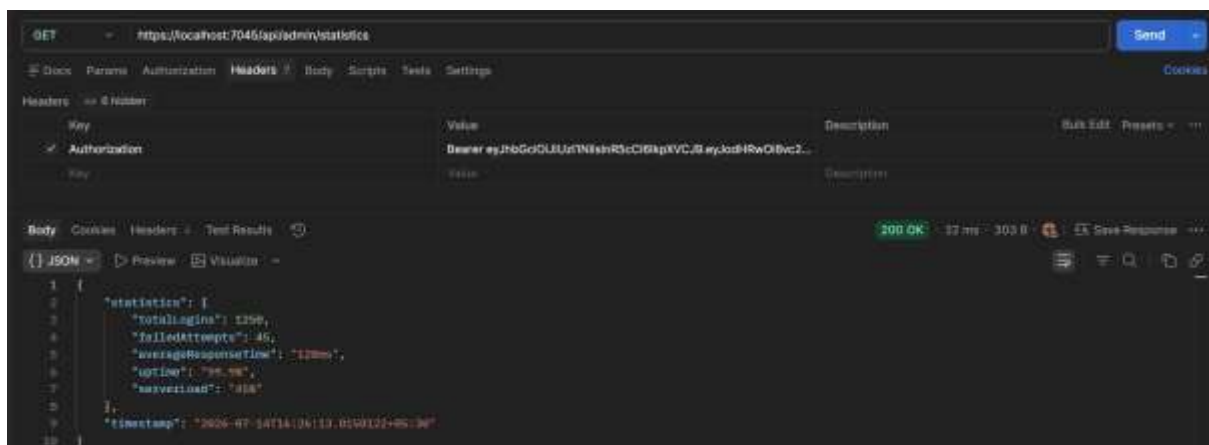
Part 1: Admin Access –



1.1 GET /api/admin/users –



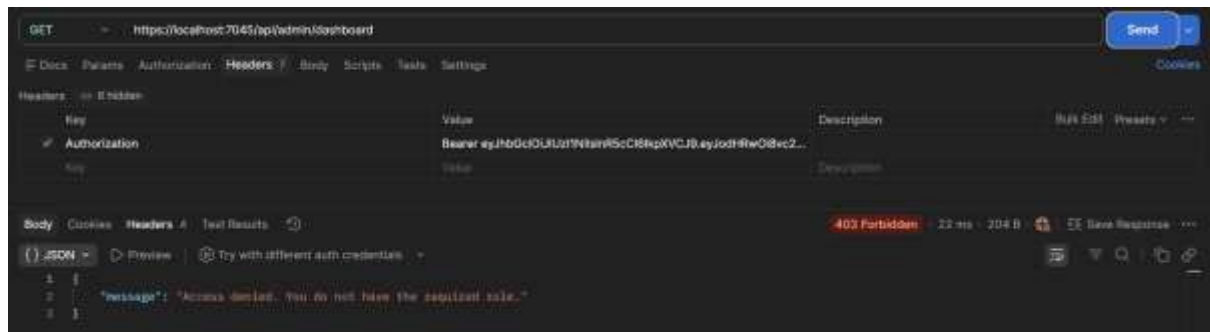
1.2 GET /api/admin/statistics –



Part 2: Regular User Access (Should Fail) –

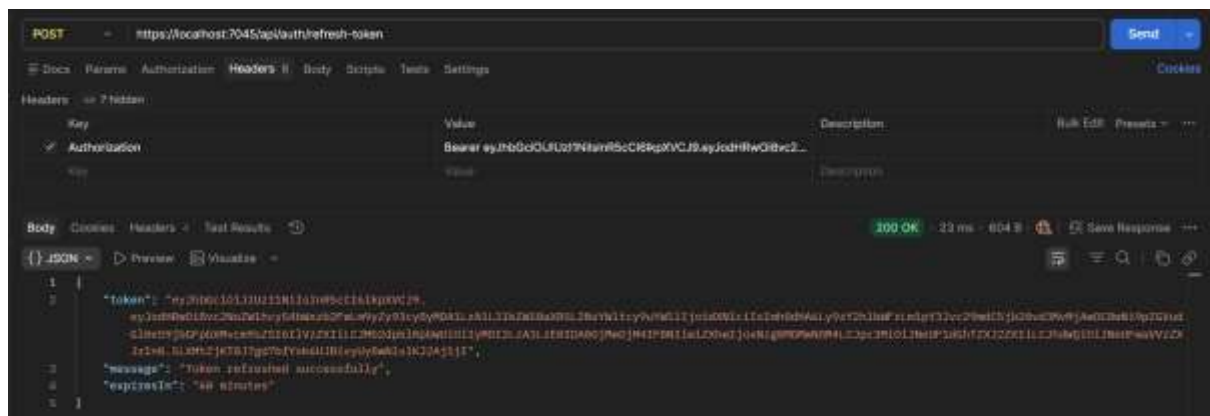
Get a new token by logging in as regular user:

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{ "username": "user", "password": "user123" }
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EXERCISE 4: TOKEN EXPIRY & REFRESH :

Test 1: Refresh Token –



Test 2: Token Expiry (Manual Test) –

- Configuration: DurationInMinutes = 1
- Token generated at: 10:45:00
- Waited: 1 minute
- Attempted access at: 10:46:05
- Expected: 401 Unauthorized
- Result:  PASSED
- Error Message: "Token has expired. Please login again."